A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 06

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 2 points below. **1** A community maintained at a stage before the climax by regular human activity (grazing, mowing, burning, coppicing) that deflects succession **1** Chalk grassland was produced by centuries of sheep grazing – without it succession to scrub and woodland resumes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **plagioclimax** → a community held back from its climax by human activity • **succession** → the gradual change in the community over time **Do not credit** • 'the climax community never changes' - it is stable only while the conditions stay the same	2

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 4 points below. **1** Grazing removes taller / dominant grasses and prevents scrub establishing (5% cover) so small plants are not shaded out **1** Trampling creates bare ground for wildflower seed germination and warm patches for invertebrates / reptiles **1** Dung creates small nutrient-rich patches supporting different plants and invertebrates – sheep eat short grasses and some bushes **1** Mowing removes everything at once including plants grazers avoid, and removes flowers / seeds before setting – burning removes vegetation but also kills invertebrates and allows scrub regrowth – abandoned plot dominated by scrub (55%) so few species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Sheep eat short grass (some breeds eat scrub) – ponies avoid flowers but eat bracken – cattle remove long rough grass **1** Choice depends on the vegetation to be controlled and the plants to be protected **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **plagioclimax** → a community held back from its climax by human activity	2
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** Removes mature heather bushes and young trees that grazers will not eat, creating open unshaded ground **1** Heat stimulates dormant heather seeds to germinate so young heather re-establishes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 47-49; AQA spec 3.1.3.3.	
02.1	**1 mark each** 1 mark for each of the 4 points below. **1** Negative **1** Positive **1** Negative **1** Positive **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4

Do not write outside the box

Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** The response to a change acts to re-establish / return the system to the previous equilibrium **1** So the change does not become self-sustaining unless a large unusual event overwhelms the negative feedback **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **negative feedback** → a change that cancels itself out / a self-correcting cycle	2
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** eg the possible release of methane hydrate at the end of the Permian era **1** Overwhelmed the negative feedback mechanisms, causing climate change and a mass extinction / a new equilibrium **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **negative feedback** → a change that cancels itself out / a self-correcting cycle	2
02.4	**1 mark each** 1 mark for each of the 2 points below. **1** Soil degradation, reduced water availability and loss of natural pest predators would reduce yields (negative feedback curtailing the activity) **1** Instead of reducing the activity, farmers use fertilisers, irrigation and pesticides, which further damage the natural systems that support food production **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **negative feedback** → a change that cancels itself out / a self-correcting cycle • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 371-372; AQA spec 3.6.1.	
03.1	**1 mark each** 1 mark for each of the 2 points below. **1** Island communities evolved from few colonising species in isolation, often without mammal predators or large herbivores **1** So indigenous species lack adaptations (eg ground-nesting birds, flightlessness) to survive introduced cats, rats, pigs, goats **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
03.2	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Competitor:** grey squirrel (North America) out-competes red squirrel - digests acorns, larger, wins nest sites – rhododendron shades ground and releases toxins preventing native regrowth **2 Predator:** American mink (escaped fur farms) reduced water vole range – cane toads in Australia – cats / rats / pigs on islands eating ground-nesting birds – Nile perch in Lake Victoria ate cichlids to extinction **2 Pathogen:** squirrel pox virus carried by grey squirrels kills red squirrels – crayfish plague carried by signal crayfish kills white-clawed crayfish – Dutch elm disease, ash dieback, sudden oak death imported on plants / soil **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	6
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Closely related introduced species (sika deer; domestic cats) cross-breed producing fertile hybrids **1** The natural gene pool is changed / diluted with genes it would not naturally contain, so the pure indigenous species may disappear **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 24-25; AQA spec 3.1.2.2.	
04.1	**1 mark each** 1 mark for each of the 3 points below. **1 MEDCs:** market demand and higher prices for the flavour of carnivorous fish (salmon, trout) justify high inputs (fishmeal, cooler climates suit them) **1 LEDCs:** herbivores / omnivores feed on natural vegetation, phytoplankton and wastes so need low inputs – warm climates suit tilapia **1** Herbivores are more energy-efficient (lower trophic level) so more food per unit area **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming • **trophic level** → the feeding level in a food chain **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3

Do not write outside the box

Question	Marking guidance	Marks
04.2	**1 mark each** 1 mark for each of the 3 points below. **1** Some genetically female (XX) fish are treated with testosterone so they become functional males producing only X-carrying sperm **1** Crossed with normal females (XX) all offspring are XX females **1** Female trout meat has a better flavour – in tilapia the reverse (all males) because males grow larger / more efficiently **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
04.3	**1 mark each** 1 mark for each of the 4 points below. **1 Salmon need high oxygen:** water sprays / weirs aerate – wastes removed to prevent deoxygenation **1** Constant circular flow keeps fish swimming the same way - fewer collisions / injuries, more muscular fish, higher stocking density **1** Artificial lighting delays maturation (salmon stop growing at maturity) so larger fish – controls smoltification **1** Cool sites (Scotland, Norway) because warm water holds less oxygen although warmth increases growth **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abiotic** → non-living / physical factors • **deoxygenation** → loss of dissolved oxygen from the water	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 352-355; AQA spec 3.5.2.3.	
05.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 C:** Waste products that are the raw materials for other processes	1
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Bacteria fix nitrogen at low temperature using enzymes and energy ultimately from sunlight (renewable) **1** The Haber process needs high temperature and pressure powered by non-renewable fossil fuels, releasing CO_2 **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2

Do not write outside the box

Question	Marking guidance	Marks
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Incineration uses high temperatures / fuel and releases pollutants – nutrients are not returned as usable raw materials **1** Decomposition at low temperature recycles nutrients as the raw materials for other natural processes (material cycle) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **incineration** → burning waste, usually with energy recovery • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2-3.6.3.	
06.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** Its mass	1
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** The 24-hour rotation period means the surface is not exposed to sunlight for long, so temperature extremes / excessive heating and cooling are reduced **1** The axis is tilted relative to the orbit so the amount of insolation at each latitude changes through the year, producing seasonal variations in conditions **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **insolation** → incoming solar radiation / energy arriving from the Sun	2
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** The distance from the Sun gives surface temperatures between 0 and 35 °C in most areas: too close and water would boil / evaporate, too far and it would freeze **1** Atmospheric pressure (produced by the mass of gases held by gravity) is high enough to stop the water boiling / evaporating away – at much lower pressure there would be no liquid water **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-7; AQA spec 3.1.1.1.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**2 marks per point** 2 marks for each of the 4 points below; award 1 mark where a point is only partly made. **2 Size:** shows whether the population is viable / how urgent action is – sets a baseline to judge success – needed to estimate carrying capacity and sustainable off-take **2 Distribution:** shows which habitats / areas are used and whether the population is fragmented into non-viable isolated groups needing corridors – identifies key areas to protect **2** Survival rate (mortality): identifies the life stage and causes of death (predation, disease, poaching) so management targets the limiting factor **2 Age structure:** many young indicates a growing population – few young / ageing population indicates breeding failure or high juvenile mortality even if adults are numerous – k-selected species need breeding adults protected **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **carrying capacity** → the largest population the area can support • **sustainable** → meeting present needs without stopping future generations meeting theirs • **limiting factor** → the factor in shortest supply that holds the rate or population back **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future • 'the population cannot grow any more' as a definition - numbers fluctuate around the carrying capacity	8
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** Direct counts from vehicles / aircraft / drones along transects, or camera traps with individual recognition **1** Sub-sample counts in known areas scaled up to the whole reserve – mark-release-recapture (Lincoln index) for smaller species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **mark-release-recapture** → estimating population size from the proportion of marked animals recaptured	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 400-402; AQA spec 3.1.2.3.5.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**Marks shared** Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2 Success:** Cactoblastis moth introduced from South America controlled prickly pear cactus in Australia – Encarsia parasitic wasps control greenhouse whitefly **up to 2 Problems:** cane toads introduced to Australia to control sugar cane beetles did not control them but ate many other species and are toxic to predators – rosy wolfsnail introduced to Polynesia to control African land snails ate endemic Partulid snails (48 species extinct) – harlequin ladybirds introduced to Europe eat native ladybirds **up to 2 Lesson:** control agents should be specialist feeders on the pest **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biological control** → using a natural predator, parasite or pathogen of the pest • **endemic** → found naturally only in that one place	4
08.2	**1 mark each** 1 mark for each of the 2 points below. **1** Release an artificial sex scent attracting males (or females) of the pest species **1** Used to monitor when the pest arrives so pesticide is timed correctly, or (in enclosed greenhouses) to trap enough of one sex to prevent breeding **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	2
08.3	**1 mark each** 1 mark for each of the 3 points below. **1** Regular low doses kill only the most sensitive bacteria so resistant strains survive and multiply **1** Agricultural use exceeds medical use, so resistance spreads **1** Resistant zoonotic bacteria (E. coli, Salmonella, Campylobacter) can transfer to humans and are harder to treat – hence reduced use of antibiotics is a sustainability goal **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainability** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3

Do not write outside the box

Question	Marking guidance	Marks
08.4	1 mark for the point below. **1** Grow nectar-rich flowering plants alongside crops – restrict pesticides harmful to bees / spray when not in flower – introduce bee hives **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 316-320; AQA spec 3.5.1.1.	
09.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 1000 / 100 000 x 100 = 1% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
09.2	**1 mark each** 1 mark for each of the 3 points below. **1** Some light is reflected by the leaves **1** Some light is transmitted (passes through the leaves without being absorbed) or is of wavelengths not absorbed **1** Much of the energy captured is used in the plant's metabolic processes and released as low energy-density heat (respiration) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** Less energy is passed to each trophic level than was received from the previous one (heat losses in respiration) **1** After four levels too little energy remains to support a further population **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **trophic level** → the feeding level in a food chain **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	2

Do not write outside the box

Question	Marking guidance	Marks
09.4	**1 mark each** 1 mark for each of the 2 points below. **1** Permanent grassland is a more sustainable land use than arable crops in these areas (soil / rainfall unsuitable for crops) **1** Grazing animals can digest grass cellulose using symbiotic bacteria in their stomachs, producing food that humans can eat **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **symbiotic** → a close relationship in which both species benefit **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
09.5	**1 mark each** 1 mark for each of the 2 points below. **1** Pigs are omnivores that can eat food wastes not wanted by people (eg cheese-making wastes, catering leftovers) **1** So edible meat is produced from material that would otherwise be wasted, without using land / crops that could feed people **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 326-328; AQA spec 3.5.1.2.	
10.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** By-catch = 5.9 + 1.9 = 7.8 t – total = 8.4 t **1** 7.8 / 8.4 x 100 = 92.9% (accept 93%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
10.2	1 mark for the point below. **1** The nets have a very small mesh size so few non-target animals can escape **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **mesh size** → net hole size, set so that young fish escape	1

Do not write outside the box

Question	Marking guidance	Marks
10.3	**1 mark each** 1 mark for each of the 3 points below. **1** They breed at an old age (skate about 11 years; Greenland shark 100-150 years) and produce few young **1** They may reach a catchable size before they are sexually mature, so fishing can remove the whole breeding population before it has bred **1** Populations recover slowly, unlike r-selected species (herring, mackerel) which breed young and produce many young – they are caught as by-catch in demersal trawls so fishing continues even though they have no commercial value **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	3

Do not write outside the box

Question	Marking guidance	Marks
10.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Impacts:** population decline of k-selected species, by-catch (immature target fish, over-quota fish, non-commercial species; drift nets - whales, dolphins, turtles; longlines - albatrosses; pair trawls - porpoises), ghost fishing, seabed damage by demersal trawls, coral and seagrass damage, dynamite fishing, food-web effects • **Catch quotas:** limit weight landed – work well for single-species pelagic shoals but in mixed fisheries over-quota fish are discarded dead so the quota fails to protect them • **Gear design:** mesh size and square-mesh escape panels let small fish escape (diagonal mesh closes under drag) – turtle exclusion devices – acoustic pingers for dolphins – curved hooks, decoys, sinkers and night setting to protect albatrosses – biodegradable / radio-tracked traps to reduce ghost fishing • **Restricted effort:** limits on boat size, engine power and days at sea – decommissioning – restricted methods - drift-net bans, demersal trawling bans on sensitive seabeds (Rockall Bank, Hatton Bank) • **Evaluation:** enforcement at sea is difficult – some methods reduce catch or raise costs – quotas need reliable stock data – effectiveness depends on species (single vs mixed shoals) and on international cooperation as fish migrate – as local stocks decline fishing moves to more distant waters • Justified conclusion weighing effectiveness of different methods **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **quota** → a legal limit on the mass or number that may be caught	9
	• **mesh size** → net hole size, set so that young fish escape • **biodegradable** → broken down by decomposers **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	
	Model answer - Level 3 (8 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Fishing's impacts go beyond depleting target species: k-selected species that breed late and produce few young collapse and recover slowly; by-catch of immature target fish, over-quota fish and non-commercial species is thrown back dead, with drift nets killing whales, dolphins and turtles, longlines drowning albatrosses and pair trawls porpoises; lost gear keeps ghost fishing; demersal trawls destroy sea fans, corals and seagrass; and removing any species alters its competitors, prey and predators.	
	Catch quotas limit the weight landed and work well for single-species pelagic shoals where fishing stops when the quota is reached, but in mixed demersal fisheries over-quota fish are discarded dead, so the quota does not protect them. Gear design tackles by-catch directly: mesh sizes and square-mesh escape panels let small fish out because, unlike diagonal mesh, they do not close under drag; turtle exclusion devices, acoustic pingers for dolphins, curved hooks, decoys, sinkers and night setting protect turtles, cetaceans and albatrosses; biodegradable and radio-tracked traps limit ghost fishing. Restricting effort - boat size, engine power, days at sea, decommissioning - reduces total pressure, and banning drift nets or demersal trawling on sensitive seabeds such as Rockall Bank protects habitats.	
	Effectiveness depends on enforcement at sea, which is difficult, on reliable stock data for quotas, on the species and fishery type, and on international cooperation because fish migrate; some measures cut catches or raise costs, and as local stocks fall fleets simply move further away. Gear modifications are the most effective per unit cost because they reduce damage without reducing legal catch; quotas and effort limits only work where they are enforced.	
	Weaker answer - Level 2 (5 marks)	
	Fishing damages the environment by overfishing, catching by-catch such as dolphins and damaging the seabed with trawling. Quotas limit how many fish can be caught so stocks can recover. Bigger mesh sizes let small fish escape so they can grow and breed. Turtle exclusion devices let turtles escape from nets. Some fishing methods can be banned in sensitive areas. No-take zones let fish breed. These methods help but they are hard to enforce at sea and some fishermen ignore them.	
	What makes the difference	
	• Level 2 covers impacts and methods soundly but without the mechanisms (why square mesh works, why quotas fail in mixed fisheries).	
	• Level 3 evaluates each method on its own terms and then compares them; Level 2's evaluation is a single general sentence.	
	• Named examples (Rockall Bank, albatross measures, ghost-fishing controls) give Level 3 the range the top band expects.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 341-345; AQA spec 3.5.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Energy flow:** autotrophs and heterotrophs – losses by reflection, transmission, respiration – 100 000 to 1 000 units – few food chains beyond four levels – plant food most productive – grazing animals where crops impossible – pigs on wastes • **Energetics:** intensive vs extensive, law of diminishing returns, energy subsidies (fertiliser, pesticides, irrigation, machinery, drying, processing, transport), energy ratios, food conversion ratios, dependence on fossil fuels • **Control of genetics:** asexual reproduction / cuttings / cloning, selective breeding and breeds, cross-breeding and hybrid vigour, AI and embryo transfer • **GM:** Roundup Ready soya, Bt crops, golden rice, omega-3 oilseed rape – advantages – risks (gene transfer, antibiotic markers, allergies, costs to LEDC farmers, loss of local diversity)	25
	• **Sustainability links:** energy dependence, trophic level of diet, feeding 12 billion people • Justified conclusion on the balance between productivity, efficiency and sustainability *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-328; AQA spec 3.5.1.2.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Ecological features:** cnidarian polyps, cilia, nematocysts, zooxanthellae symbiosis, limestone skeletons building reefs, complex structure and food web – specific abiotic requirements (temperature, light, salinity, low turbidity) • **Importance:** quarter of marine species, fisheries, medicines, carbon sink, erosion protection, tourism • **Vulnerability:** narrow abiotic tolerance means sensitivity to warming, acidification, sedimentation, nutrient pollution, oil, bleaching – slow growth so slow recovery – physical damage – overfishing and food-web effects (crown of thorns) – introduced species – dependence on mangroves and seagrass • **Conservation:** MPAs, No Take Zones, fishing controls, tourism management, protection of linked habitats, CITES • Comparison with other habitats (rainforests, mangroves, Antarctica) where relevant • Evaluation of local vs global threats and justified conclusion	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-62; AQA spec 3.1.2.3.4.	

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